Linear Algebra

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Source Code & Contributions
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Contents

Preface v

1 Vector Spaces 1
- 1.1 Definitions . . . 1
- 1.2 Bases . . . 1
- 1.3 Dimension of a Vector Space . . . 1
- 1.4 Sums and Direct Sums . . . 1

2 Matrices 2
- 2.1 The Space of Matrices . . . 2
- 2.2 Linear Equations . . . 2
- 2.3 Multiplication of Matrices . . . 2

3 Linear Mappings 3
- 3.1 Mappings . . . 3
- 3.2 Linear Mappings . . . 3
- 3.3 The Kernel and Image of a Linear Map . . . 3
- 3.4 Composition and Inverse of Linear Mappings . . . 3
- 3.5 Geometric Applications . . . 3

4 Linear Maps and Matrices 4
- 4.1 The Linear Map Associated with a Matrix . . . 4
- 4.2 The Matrix Associated with a Linear Map . . . 4
- 4.3 Bases, Matrices, and Linear Maps . . . 4

5 Scalar Products and Orthogonality 5
- 5.1 Scalar Products . . . 5
- 5.2 Orthogonal Bases, Positive Definite Case . . . 5
- 5.3 Application to Linear Equations; the Rank . . . 5
- 5.4 Bilinear Maps and Matrices . . . 5
- 5.5 General Orthogonal Bases . . . 5
- 5.6 The Dual Space and Scalar Products . . . 5
- 5.7 Quadratic Forms . . . 5

5.8 Sylvester's Theorem 5

6 Determinants 6
6.1 Determinants of Order 2 6
6.2 Existence of Determinants 6
6.3 Additional Properties of Determinants 6
6.4 Cramer's Rule 6
6.5 Triangulation of a Matrix by Column Operations 6
6.6 Permutations 6
6.7 Expansion Formula and Uniqueness of Determinants 6
6.8 Inverse of a Matrix 6
6.9 The Rank of a Matrix and Subdeterminants 6

7 Symmetric, Hermitian, and Unitary Operators 7
7.1 Symmetric Operators 7
7.2 Hermitian Operators 7
7.3 Unitary Operators 7

8 Eigenvectors and Eigenvalues 8
8.1 Eigenvectors and Eigenvalues 8
8.2 The Characteristic Polynomial 8
8.3 Eigenvalues and Eigenvectors of Symmetric Matrices 8
8.4 Diagonalizations of a Symmetric Linear Map 8
8.5 The Hermitian Case 8
8.6 Unitary Operators 8

9 Polynomials and Matrices 9
9.1 Polynomials 9
9.2 Polynomials of Matrices and Linear Maps 9

10 Triangulation of Matrices and Linear Maps 10
10.1 Existence of Triangulation 10
10.2 Theorem of Hamilton-Cayley 10
10.3 Diagonalization of Unitary Maps 10

11 Polynomials and Primary Decomposition 11
11.1 The Euclidean Algorithm 11
11.2 Greatest Common Divisor 11
11.3 Unique Factorization 11
11.4 Application to the Decomposition of a Vector Space 11
11.5 Schur's Lemma 11
11.6 The Jordan Normal Form 11

12 Convex Sets **12**
12.1 Definitions 12
12.2 Separating Hyperplanes 12
12.3 Extreme Points and Supporting Hyperplanes 12
12.4 The Krein-Milman Theorem 12

Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.34.0 and Mathlib v4.34.0 (revision `5ed296525643`).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.

Chapter 1

Vector Spaces

1.1 Definitions

1.2 Bases

1.3 Dimension of a Vector Space

1.4 Sums and Direct Sums

Chapter 2

Matrices

2.1 The Space of Matrices

2.2 Linear Equations

2.3 Multiplication of Matrices

Chapter 3

Linear Mappings

3.1 Mappings

3.2 Linear Mappings

3.3 The Kernel and Image of a Linear Map

3.4 Composition and Inverse of Linear Mappings

3.5 Geometric Applications

Chapter 4

Linear Maps and Matrices

4.1 The Linear Map Associated with a Matrix

4.2 The Matrix Associated with a Linear Map

4.3 Bases, Matrices, and Linear Maps

Chapter 5

Scalar Products and Orthogonality

5.1 Scalar Products

5.2 Orthogonal Bases, Positive Definite Case

5.3 Application to Linear Equations; the Rank

5.4 Bilinear Maps and Matrices

5.5 General Orthogonal Bases

5.6 The Dual Space and Scalar Products

5.7 Quadratic Forms

5.8 Sylvester's Theorem

Chapter 6

Determinants

6.1 Determinants of Order 2

6.2 Existence of Determinants

6.3 Additional Properties of Determinants

6.4 Cramer's Rule

6.5 Triangulation of a Matrix by Column Operations

6.6 Permutations

6.7 Expansion Formula and Uniqueness of Determinants

6.8 Inverse of a Matrix

6.9 The Rank of a Matrix and Subdeterminants

Chapter 7

Symmetric, Hermitian, and Unitary Operators

7.1 Symmetric Operators

7.2 Hermitian Operators

7.3 Unitary Operators

Chapter 8

Eigenvectors and Eigenvalues

8.1 Eigenvectors and Eigenvalues

8.2 The Characteristic Polynomial

8.3 Eigenvalues and Eigenvectors of Symmetric Matrices

8.4 Diagonalizations of a Symmetric Linear Map

8.5 The Hermitian Case

8.6 Unitary Operators

Chapter 9

Polynomials and Matrices

9.1 Polynomials

9.2 Polynomials of Matrices and Linear Maps

Chapter 10

Triangulation of Matrices and Linear Maps

10.1 Existence of Triangulation

10.2 Theorem of Hamilton-Cayley

10.3 Diagonalization of Unitary Maps

Chapter 11

Polynomials and Primary Decomposition

11.1 The Euclidean Algorithm

11.2 Greatest Common Divisor

11.3 Unique Factorization

11.4 Application to the Decomposition of a Vector Space

11.5 Schur's Lemma

11.6 The Jordan Normal Form

Chapter 12

Convex Sets

12.1 Definitions

12.2 Separating Hyperplanes

12.3 Extreme Points and Supporting Hyperplanes

12.4 The Krein-Milman Theorem

Bibliography

[1] Serge Lang, *Linear Algebra*, Springer Nature.